

Multidisciplinary Engineer combining mathematical rigor with high-performance systems architecture. My expertise spans the full computing spectrum: from designing proprietary **Quantitative Inference Engines** (Colt) and strictly-typed **Compilers** (Σ), to building low-latency **Game Frameworks** (Piper) and distributed **Web Systems**. I specialize in solving non-stationary problems where standard tooling fails.

ENGINEERING & RESEARCH

Colt v1.0: Quantitative Inference Engine 2025

- PyTorch • Stochastic Calculus • Reinforcement Logic • Monte Carlo
- Architecture: Designed "MoRT" (Mixture-of-RecurSIONs Transformer), a proprietary architecture that solves the Stability-Plasticity Dilemma by using Token-Choice Routing to dynamically allocate computational depth based on market entropy.
 - Optimization: Replaced standard predictive loss functions (MSE/Cross-Entropy) with a custom **Differentiable Sharpe Ratio**, aligning gradient descent directly with portfolio risk-adjusted returns.
 - Results: Audited on 5,000 out-of-sample bars (2025), achieving an **SQN of 6.52** and **Sharpe Ratio of 8.90** on USD/JPY. Demonstrated "Crisis Alpha" behavior by decoupling from -25% market drawdowns.

Sigma (Σ) Language & Compiler 2024 - Present

- Rust • LLVM • Python FFI • Graph Reduction
- Design: Architected a strictly typed, purely functional DSL for mathematical logic with Hindley-Milner type inference, bridging the gap between Haskell's safety and Python's usability.
 - Runtime: Built a custom **Rust Bytecode VM** that implements Graph Reduction semantics for lazy evaluation. Engineered a "Zero-Copy" FFI bridge (via PyO3) to pass NumPy arrays directly to the VM without serialization overhead.
 - Performance: Implemented SIMD-optimized vector operations and Copy-on-Write (CoW) memory management for high-throughput tensor processing.

Universal Inventory Manager (SaaS) 2024

- C# • .NET 8 • Blazor Server • SignalR • SQL
- Concurrency: Engineered a real-time, multi-tenant inventory system using WebSocket (SignalR) pipelines to handle concurrent updates across distributed teams with sub-100ms latency.
 - Security: Implemented a robust RBAC (Role-Based Access Control) system with OAuth integration and a custom ID generation logic to ensure deterministic consistency across sharded databases.

Akshara (System Utility) 2024

- Haskell • C (FFI) • Windows API • Algorithm Design
- Architecture: Developed the spiritual successor to Avro Keyboard, decoupling the phonetic engine into a standalone **Haskell** library ("The Brain") that communicates with OS-level hooks via a C Foreign Function Interface ("The Body").
 - Impact: Achieved platform agnosticism (Windows/Linux/macOS support) and significantly reduced input latency compared to legacy C++ implementations through functional state management.

Piper Game Framework 2023

- C99 • OpenGL • Memory Management • Single-Header
- Constraints: Engineered a **Single-Header** 2D game framework (<1.2MB) with zero external dependencies. Embeds windowing (RGFW), audio mixing, and vector math into a single portable unit.
 - Graphics: Wrote a custom software renderer from scratch, including a matrix stack for transformations, bilinear texture filtering, and a batching system for high-performance sprite rendering.

THE POLYMATH STACK

Core Languages	Rust, Haskell, C/C++, C#, Python, TypeScript, GLSL
Quant Finance	Stochastic Calculus, Time-Series Analysis, Market Microstructure, Monte Carlo
Systems	Compiler Design (LLVM), OpenGL, Graph Reduction, Memory Management
Web / Cloud	.NET 8, Blazor, SignalR, Docker, CI/CD Pipelines, SQL

EDUCATION